

Single Source Shortest Path (SSSP)

The algo we will discuss will solve SSSP problem

Given source s output a shortest path from s to every other vertex

Negative-weight edges

- if there is a cycle in G which is reachable on path from s to t and the sum of weights of the edges in the cycle is $-ve$, then we get to t from s with a path with lower weight than the starting point
- Bellman-Ford detects negative weight cycle

Notations

$w(u, v)$ — weight for an edge (u, v)
 $(u, v) \rightarrow$ distance from u to v (i.e. total weight)
 $(u, v) = \infty$ when there is no path from u to v

For each vertex we will maintain a guess for distance (weight) from source ($v.d$)

$v.T$ — Predecessor

Framework of Bellman-ford

- Initialize a guess $v.d$ for distance from source S . $S.d = 0$ and $v.d = \infty$ for all other vertices
- update the guesses
 - = if there is an edge $u \rightarrow v$ and our guess for the distance from S to v is $>$ our guess from S to $u + w(u, v)$ then we can improve.

Relax(u, v, w)

if $v.d > u.d + w(u, v)$

$v.d = u.d + w(u, v)$

$v.\pi = u$

BellmanFord(G, S)

for each vertex $v \in G$

$v.d = \infty$

$v.\pi = Nil$

$S.d = 0$

for $i = 1$ to $|V| - 1$

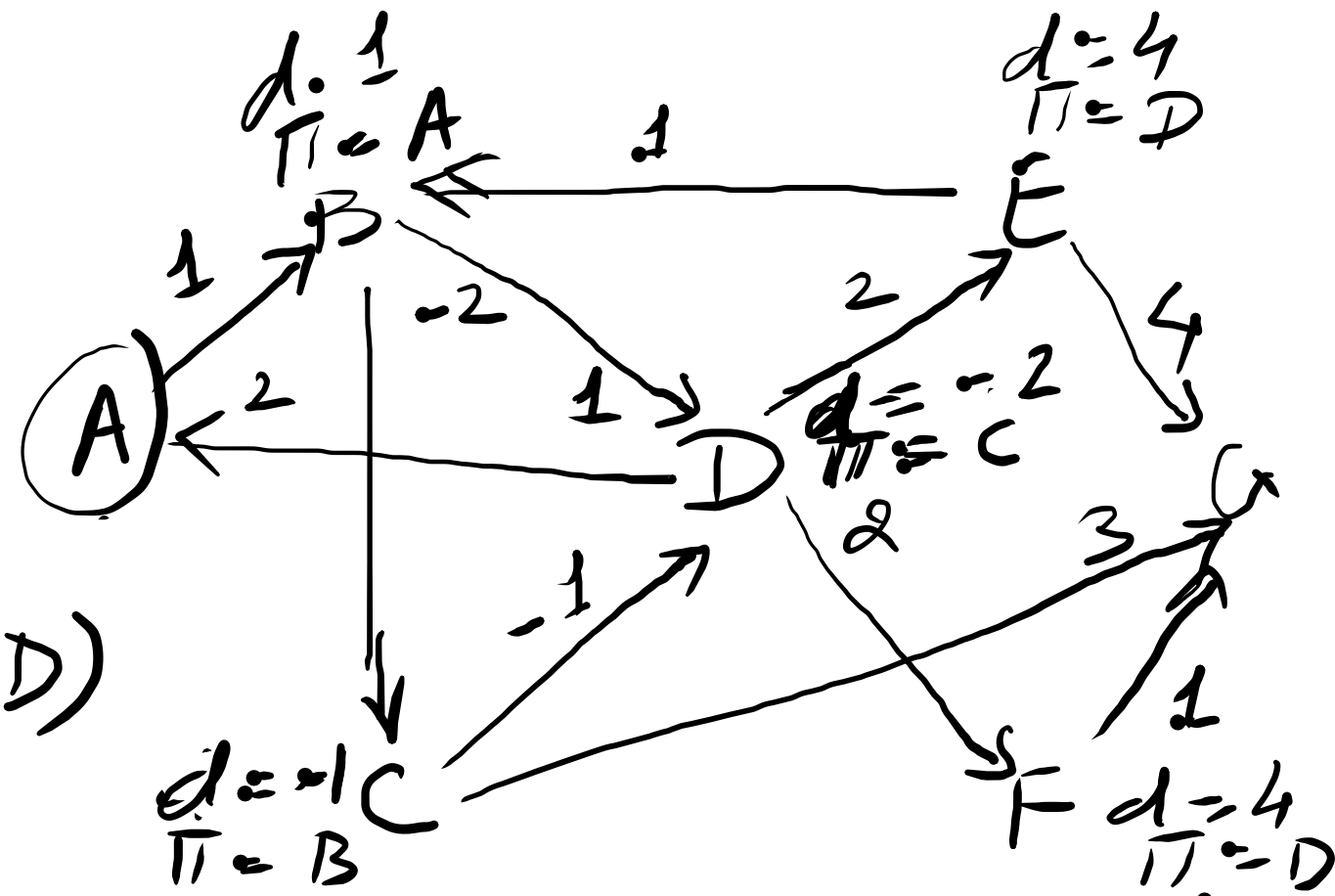
for each edge $u \rightarrow v$ in $G.E$

Relax(u, v, w)

(B, C) (B, D) (A, B) (D, A) (D, F) (C, D) (C, G)

(E, B) (E, G) (F, G)

(D, E)



$A.d = 0$

$A.\pi = Nil$

$B.d = 1$

$B.\pi = A$

$C.d = -1$

$C.\pi = B$

$D.d = -2$

$D.\pi = C$

$E.d = 4$

$E.\pi = D$

$F.d = 4$

$F.\pi = D$

$G.d = \infty$

$G.\pi = Nil$